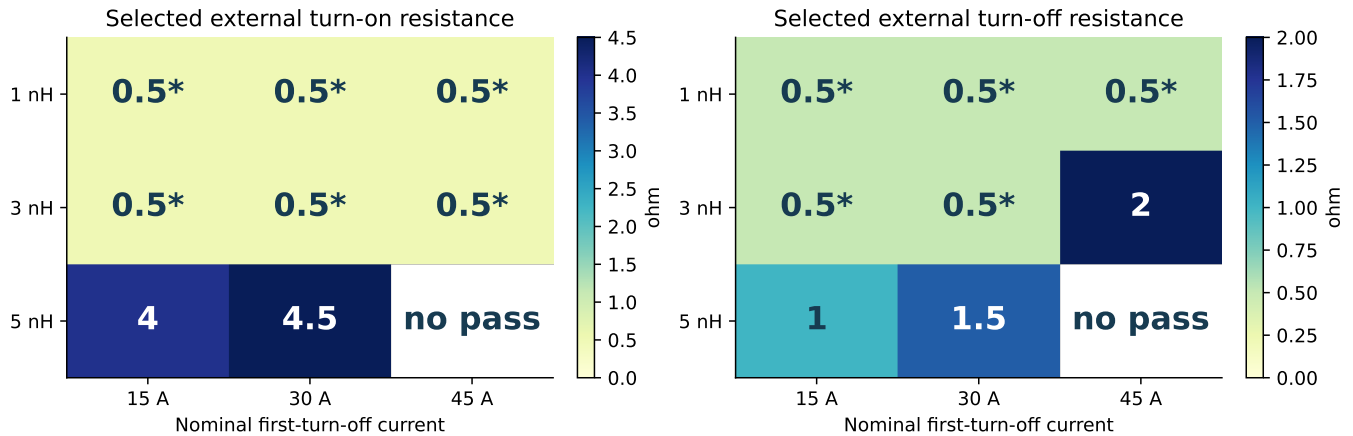


Minimum tested gate resistance

Constraint: maximum DUT $V_{DS} < 100$ V over the entire DPT.

17 September 2026



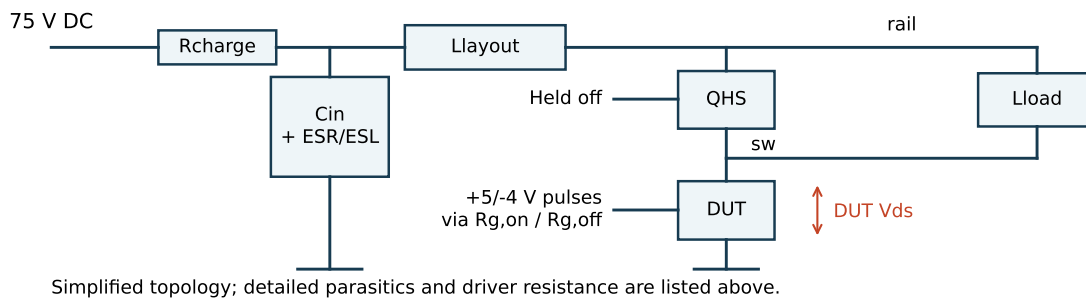
L_{layout} nH	Setpoint A	Measured I_{off1}/I_{off2} A	External R_{on}/R_{off} Ω	Worst checked peak V	Margin V
1	15	15.02/18.45	0.5/0.5*	88.47	11.53
1	30	30.02/33.37	0.5/0.5*	88.21	11.79
1	45	45.02/48.30	0.5/0.5*	91.80	8.20
3	15	15.02/18.46	0.5/0.5*	89.66	10.34
3	30	30.02/33.38	0.5/0.5*	97.08	2.92
3	45	45.02/48.32	0.5/2*	96.22	3.78
5	15	14.97/18.34	4/1	99.66	0.34
5	30	29.96/33.26	4.5/1.5	99.63	0.37
5	45	—	no pass	—	—

Selection: minimize $R_{on} + R_{off}$ on a 0.5–12 Ω grid in 0.5 Ω steps; ties favor lower R_{on} . All lower-sum grid pairs are checked before selection. Runs at 0.5, 0.25 and 0.125 ns must all pass. This optimizes the stated resistor objective, not measured switching time or loss. *Search floor reached; smaller values were not tested.

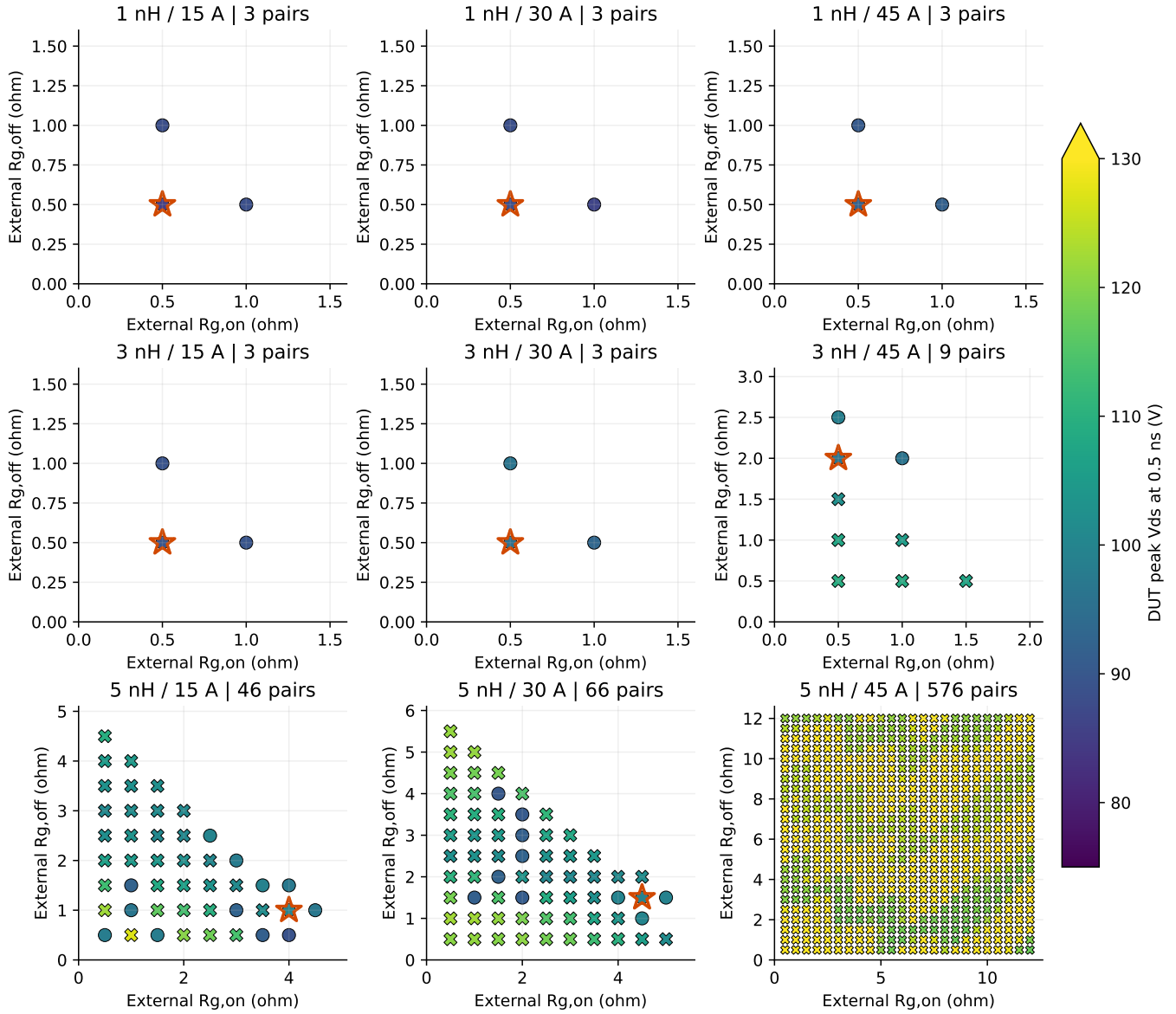
Fixed setup: EPC2361, 75 V DC link, 100 °C, +5/-4 V gate rails; $C_{in} = 470 \mu\text{F}$, ESR=5 m Ω , ESL=0.2 nH; $L_{load} = 20 \mu\text{H}$, DCR=20 m Ω ; loop R=5 m Ω , CSI=0.1 nH per device. Driver adds 0.7/0.4 Ω to the listed external on/off resistors. Upper gate held off through 0.9 Ω .

Voltage margin: peaks near 100 V are boundary candidates, with insufficient demonstrated allowance for hardware tolerances. The simple $di/dt = (100 - 75)/L$ budget is 25, 8.3 and 5 A/ns at 1, 3 and 5 nH; additional inductance and ringing reduce this budget.

Current definition: first pulse calibrated once per cell for the first turn-off setpoint, then held fixed during resistor search. The second 1 μs pulse raises current further; its turn-off is included in the voltage constraint. Off gap=1 μs ; command edges=2 ns.

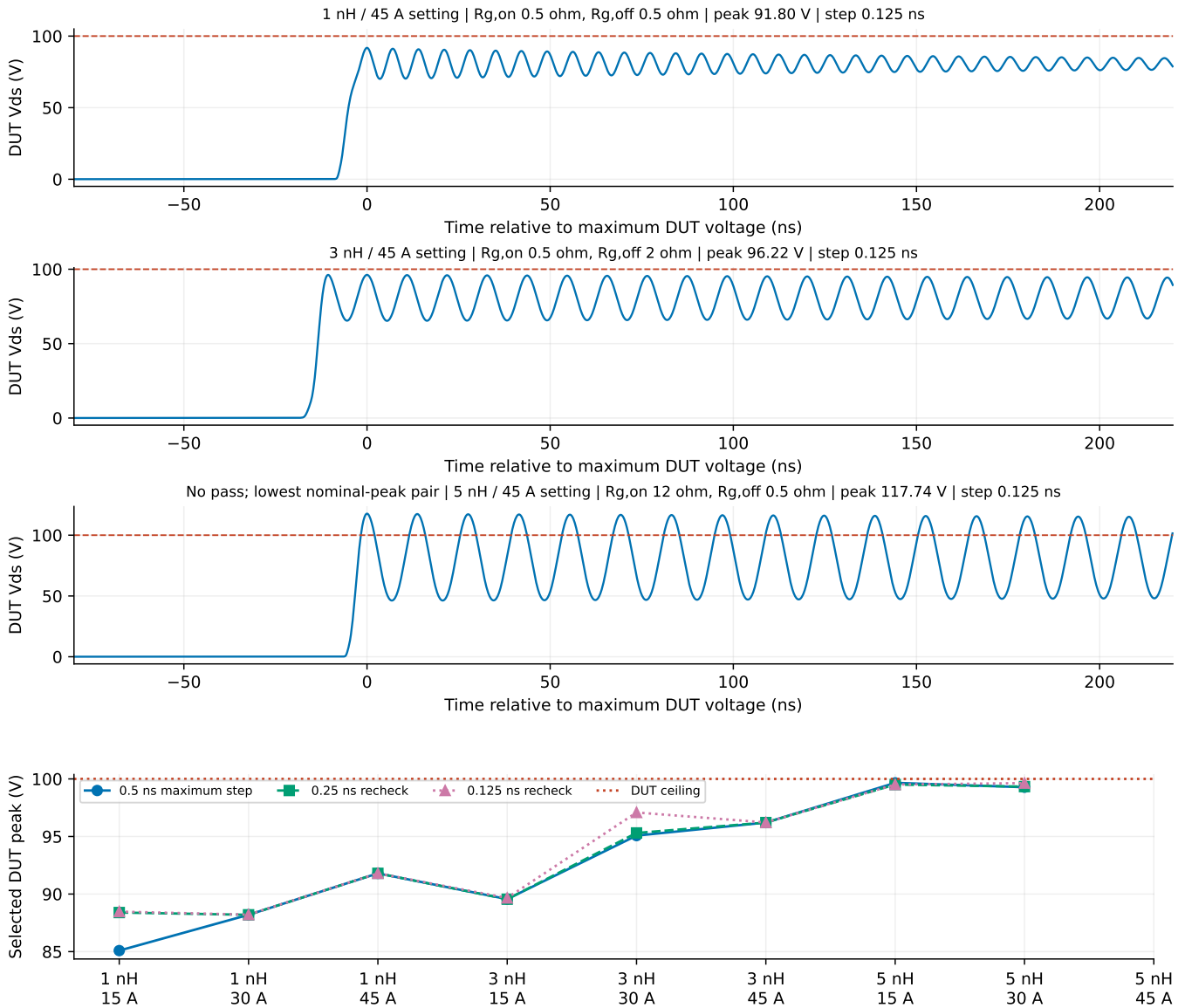


Tested resistance pairs



Circle: nominal-step pass. X: DUT peak ≥ 100 V. Orange star: selected and checked at both finer steps. Red +: unresolved run. Blank locations were not tested. Colour is the DUT voltage only; values above 130 V saturate the scale. Additional neighbours show local sensitivity and do not imply monotonic behaviour. No pass means no verified pair was found; unresolved runs cannot be classified as voltage failures. 750 search/verification/neighbour runs; 0 unresolved. Separate calibration runs are retained with the data.

Selected-pair voltage checks



Waveform timesteps are shown in the panel titles. For a case with no pass, the lowest nominal-peak pair is shown (finer-step recheck when available) and omitted from the selected-pair comparison. **Scope:** smallest tested pairs for this model and test sequence. Full numerical convergence and bus, temperature, device or parasitic tolerance envelopes were not qualified. The upper-device voltage was excluded as requested; that does not establish its physical safety. The -4 V off setting is at the EPC2361 negative gate limit and has no undershoot allowance. These are DUT-voltage-pass candidates, not half-bridge-safe gate-drive prescriptions.

Reproduce: `python search_gate_limits.py`; `python build_gate_report.py`. Gear solver, $\text{reitol}=10^{-4}$, $\text{abstol}=10^{-9}$ A, $\text{vntol}=10^{-6}$ V; no waveform compression. Data: `dut_gate_search/trials.csv`, per-case JSON/netlists/raw files and model hash. Existing GUI unchanged.

Rating reference: EPC2361 datasheet (July 20, 2026), p.1: 100 V continuous; 120 V repetitive transient only for duty factor $\leq 1\%$. This study uses 100 V. https://epc-co.com/epc/Portals/0/epc/documents/datasheets/EPC2361_datasheet.pdf